

Katherine Titterton

San Francisco, CA · macrofluidic.me · github.com/kat-titter · X @KatTitterton

TL;DR Neuroscientist-engineer at the seam of imaging, signal extraction, and machine learning. 8+ years building the instruments, datasets and models that turn biological signal into measurement: voltage-clamp recordings, expansion microscopy, PyTorch segmentation over 100,000+ compound image sets. Founding technical hire with product taste; artist in residence.



MACROFLUIDIC.ME/RESUME

100,000+

compounds in two of the largest
Cell Painting datasets

U-Net

segmentation in daily use 1+
year; SAM benchmarked

~10×

more sensitive than standard
assays, from images

8 papers

peer-reviewed imaging
methods, incl. Nat Commun ×2

EXPERIENCE

[n] = publication number on page 2

Founding Member of the Technical Staff · Axiom Bio

2023–present · San Francisco, CA

- Built and maintained PyTorch U-Net segmentation models over 100,000+ compound Cell Painting images; daily use 1+ year; SAM benchmarked.
- Oversaw assay development, QC and data generation for two of the world's largest Cell Painting datasets (training data for in silico toxicology).
- Invented two image-based bioactivity-detection methods ~10× more sensitive than conventional assays; drove product direction. Artist in Residence.

Nat Commun 2026 [4] · Cell Syst 2026 [3]

Associate Scientist III · Insitro, Arrayed Screening, Neuroscience

2022–2023 · South San Francisco, CA

- Scaled high-content phenotypic assays to automated high-throughput small-molecule screens (TSC, ALS; BioTek, High-Res platforms).
- Designed and validated a label-free phase-contrast timecourse imaging modality and Python analysis suite to track neuron growth.

Technical Associate II · MIT, Synthetic Neurobiology Group

2021–2022 · Cambridge, MA

- Imaged 30+ proteins at nanoscale by optimizing multiplexed expansion revealing (mExR) for 7+ IHC rounds on 16× expanded tissue.
- Built and ran the image-analysis pipelines (ImageJ/Fiji, MATLAB) behind the published method.

Nat Commun 2024 [6]

Associate Scientist II · AbbVie, Foundational Neuroscience Center

2018–2021 · Cambridge, MA

- Built primary murine and hiPSC neuron tau-aggregation high-content assays (Revvity Opera, Harmony) to rank Alzheimer's lead candidates.
- Designed microfluidic neuron-culture devices to model tau propagation; authored R pipelines for reproducible dose-response profiling.

Mol Neurodegener Adv 2026 [1] · iScience 2026 [2] · J Biol Chem 2025 [5] · iScience 2022 [7]

Research Technician I · Massachusetts General Hospital

2017–2018 · Boston, MA

- Recorded and analyzed voltage-clamp currents from GABA-A-receptor *Xenopus* oocytes to resolve mechanisms of general anesthetics.

Eur J Med Chem 2019 [8]

PROJECTS

Bare Metal · Solo generative-art exhibit, Human Tech Week, SF; concept, fabrication and post-hoc analysis (macrofluidic.me/blog)

2025

Vivarium 2.0 · Designed and built an automated poison-dart-frog habitat: misting control, motion-triggered live video

2024

OpenTransformers · LLM code-generation companion that lets wet-lab scientists drive the OT-2 robot; 2nd place, BioHack NYC

2023

EDUCATION

Harvard University · Extension School, graduate coursework

2017–2020

Graduate coursework while employed full-time: Python programming; image processing & computer vision; social medicine.

B.A., Colgate University · *cum laude*

2013–2017

Cellular Neuroscience major, Mathematical Systems Biology minor; biophysics, neuroethology. Co-captain, Division I Track & Field.

SKILLS & TOOLS

ML & DATA

PyTorch · deep-learning segmentation (U-Net; SAM evaluated) · image embeddings: clustering, feature derivation · Python (NumPy, SciPy, scikit-learn, pandas) · multivariate statistics · LLM codegen (Claude Code, Cursor)

HARDWARE & AUTOMATION

Liquid handlers & schedulers: OpenTrons, Integra, BioTek, Labcyte Echo, High-Res Biosolutions, Green Button Go · microfluidic devices · SOP-driven validation & QC

IMAGING & SIGNAL

High-content & confocal microscopy (Opera Phenix, Operetta) · expansion microscopy · phase-contrast timecourse · voltage-clamp electrophysiology · ImageJ/Fiji macros, MATLAB · curve fitting (R, Python)

NEURO WET LAB

Primary murine & hiPSC neuron culture · ICC/IHC · stereotaxic surgery, tissue microdissection · assay development & HT/HC screening

CERTIFICATIONS

FII Level 1 Freediving (Oct 2019) · Molchanov L2 Freediving (Oct 2025) · SII Shark Ecology Specialty (Apr 2020)

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PEER-REVIEWED PUBLICATIONS

8 papers · newest first

- [1] Yanamandra K, Romanul N, Patel H, Sanders H, Chang R, Darcy A, Brown N, Kane C, Wu HY, Jagtap K, Chancellor S, Morozova O, Xi M, Hogan C, **Titterton K**, Wu JW, Manos J, Karran E, Dellovade T, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv.* 2026;2(1):43. doi:10.1186/s44477-026-00049-6 **ABBVIE**
- [2] Sebastian RM, Kupp R, Nanavati D, Parfentev I, Krzystek TJ, Romanul N, Yanamandra K, Preiss CN, Manos JD, Wu HY, Volkova A, Crescenti S, **Titterton K**, Welker AM, Townsend M, Karran E, Langlois X, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience.* 2026;29(7):116362. doi:10.1016/j.isci.2026.116362 PMID 42325569 · PMC13276770 **ABBVIE**
- [3] Ewald JD, **Titterton KL**, Bäuerle A, Beatson A, Boiko DA, Cabrera ÁA, Cheah J, Cimini BA, Gorissen BL, Harrill J, Jones TR, Karczewski KJ, Crute CE, Rouquie D, Seal S, Weisbart E, White B, Carpenter AE, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst.* 2026;17(5):101566. doi:10.1016/j.cels.2026.101566 PMID 41903531 **AXIOM BIO**
- [4] Seal S, Dee W, Shah A, Cerisier N, Zhang A, Miglietta E, **Titterton K**, Cabrera ÁA, Boiko D, Beatson A, Slabaugh G, Taboureaux O, Carreras Puigvert J, Singh S, Spjuth O, Bender A, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun.* 2026;17(1). doi:10.1038/s41467-026-68725-5 PMID 41651839 · PMC12988037 **AXIOM BIO**
- [5] Wu JW, **Titterton K**, Sebastian R, Romanul N, Yanamandra K, Mastis B, Chang R, Khasnavis S, Kwon T, Chai Y, Manos JD, Preiss CN, Hu M, Welker AM, Liao F, Dellovade T, Karran E, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem.* 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853 PMID 41135680 · PMC12670552 **ABBVIE**
- [6] Kang J, Schroeder ME, Lee Y, Kapoor C, Yu E, Tarr TB, **Titterton K**, Zeng M, Park D, Niederst E, Wei D, Feng G, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun.* 2024;15(1):9722. doi:10.1038/s41467-024-53729-w PMID 39521775 · PMC11550395 **MIT**
- [7] Manos JD, Preiss CN, Venkat N, Tamm J, Reinhardt P, Kwon T, Wu J, Winter AD, Jahn TR, Yanamandra K, **Titterton K**, Karran E, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience.* 2022;25(1):103658. doi:10.1016/j.isci.2021.103658 PMID 35072001 · PMC8761709 **ABBVIE**
- [8] Wu B, Jayakar SS, Zhou X, **Titterton K**, Chiara DC, Szabo AL, Savechenkov PY, Kent DE, Cohen JB, Forman SA, Miller KW, Bruzik KS. Inhibitable photolabeling by neurosteroid diazirine analog in the β 3-subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem.* 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019 PMID 30544077 · PMC6912599 **MGH**

CONFERENCE PRESENTATIONS & POSTERS

- [1] **Titterton K**, Boiko D, Cabrera ÁA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. "Accurate and interpretable in silico clinical risk assessment for drug-induced liver injury (DILI) from molecular structure." *Society of Toxicology (SOT) Annual Meeting*, 2025.
- [2] **Titterton K**, et al. "Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation." *Society for Neuroscience (SfN) Global Connectome Virtual Meeting*, January 2021.
- [3] **Titterton K**, et al. "High content analysis (HCA) of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression in vitro." *SBI2 High Content Meeting*, September 2019, Harvard Medical School, Boston, MA.

Earlier research presentations

- "Investigating RNAi as a potential therapeutic for Huntington's Disease in a transgenic mHTT145Q zebra finch (*Taeniopygia guttata*) model." Colgate University Neuroscience Thesis Poster Presentations, 2017.
- "Atopy and helminth infection in Ethiopian schoolchildren." Colgate Summer Research Poster Symposium, 2016.
- "PDE2 regulation of β -arrestin signaling." Systems Biology Center New York (SBCNY) Summer Trainee Poster Symposium, Icahn School of Medicine at Mount Sinai, 2015.